GATE

Computer Science & IT

Computer Networks
Switching and Routing

DPP: 1

Q1 Identify correct statement regarding 'Link State' routing algorithm.
(A) Dijkstra Algorithm executed by any one router
(B) Dijkstra Algorithm executed by a particular routers
(C) Dijkstra Algorithm executed by all the routers
(D) No need to run Dijkstra Algorithm explicitly.

Q2 Identify correct statement regarding 'Distance Vector' routing algorithm.
(A) Does not suffer with routing loop
(B) High packets overhead per network
(C) Every router broadcast its 'Distance Vector'
(D) None of the above

Q3 Consider a network of five routers : P, Q, R, S and T, uses 'Distance Vector' routing algorithm. Every router maintain distance to other router in the above mentioned routers sequence. Router Q is directly connected with router P and S with distance 4 and 2 respectively. If distance vector of P : [0, 4, 3, 5, 2] and distance vector of S : [6, 2, 4, 0, 3] are arrived at router Q, then what should be router Q updated distance vector?
(A) [4, 0, 6, 2, 6]
(B) [4, 0, 7, 2, 5]
(C) [4, 0, 6, 2, 5]
(D) [4, 0, 7, 2, 6]

Q4 Consider a network of three routers A, B, and C, where B is connected to A and C with distance 4 and 5 respectively.
Identify correct distance vector of router B after one iteration (round of update) of the distance vector routing algorithm?
(A) A = 5, B = 0, C = 4
(B) A = 4, B = 0, C = ∞
(C) A = ∞, B = 0, C = 5
(D) A = 4, B = 0, C = 5

Q5 Identify correct statements regarding 'Distance Vector' and 'Link State' routing algorithm.
(A) RIP uses 'Distance Vector' routing algorithm
(B) RIP uses 'Link State' routing algorithm
(C) OSPF uses 'Distance Vector' routing algorithm
(D) OSPF uses 'Link State' routing algorithm

Q6 Identify INCORRECT statement(s) regarding 'RIP' and 'OSPF' routing protocol.
(A) RIP uses TCP (B) RIP uses UDP
(C) OSPF uses TCP (D) OSPF uses UDP

Q7 Which switching technique lead inefficient utilization of network resources?
(A) Circuit Switching
(B) Packet Switching
(C) Virtual-circuit Switching
(D) Message Switching

Q8 Which switching techniques provide "connection-less" services?
(A) Circuit Switching
(B) Packet Switching
(C) Virtual-circuit Switching
(D) None of the above

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Answer Key

Q1	(C)	Q5	(A, D)
Q2	(D)	Q6	(A, C, D)
Q3	(C)	Q7	(A)
Q4	(D)	Q8	(B)

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Hints & Solutions

Q1 Text Solution:

Dijkstra Algorithm executed by all the routers

Q2 Text Solution:

None of the above

Q3 Text Solution:

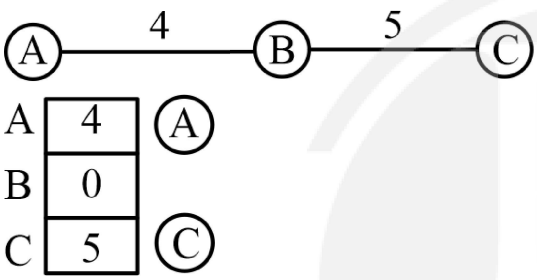
P : 4 [0, 4, 3, 5, 2]

S : 2 [6, 2, 4, 0, 3]

Q. P Q R S T

4, 0, 6, 2, 5

Q4 Text Solution:



Q5 Text Solution:

(a) RIP uses 'Distance Vector' routing algorithm **True**

(b) RIP uses 'Link State' routing algorithm **False**

(c) OSPF uses 'Distance Vector' routing algorithm **False**

(d) OSPF uses 'Link State' routing algorithm **True**

Q6 Text Solution:

(a) RIP uses TCP **False**

(b) RIP uses UDP **True**

(c) OSPF uses TCP **False**

(d) OSPF uses UDP **False**

RIP → UDP → IP

OSPF → IP

Q7 Text Solution:

Circuit Switching → Path reservation, leads efficient utilization of network resources.

Q8 Text Solution:

(a) Circuit Switching → Provide connection oriented serices

(b) Packet Switching → Provide connection less services

(c) Virtual-circuit Switching → Connection oriented packet switching

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